

CS 511: Applied Probability

Course Code	CS 511
Course Name	Applied Probability
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MTech (CSE), MS, PhD
Mutual Exclusion	EE534, MA524
Approval	44th BoA (Nov 2021)

Course Contents

- **Module I:** Sigma field. Review of axiomatic probability, conditional probability, and independence, Bayes rule and applications. [3 Lectures]
- **Module II:** Recap of random variables, discrete and continuous random variables, and functions of random variables. [2 Lectures]
- **Module III:** Joint, marginal, and conditional distribution, Covariance and correlation, Multinomial, Multivariate Normal, Conditional Expectations. [2 Lectures]
- **Module IV:** Probability generating function, moment generating function and characteristic functions properties and applications. [3 Lectures]
- **Module V:** Markov chains, classification of states and chains, stationary distribution and limit theorem, Poisson process. Application of Markov Chain in Page Rank, text summarization etc. [4 Lectures]
- **Module VI:** The convergence of random variables basic results, inequalities (Markov and Chebyshev), the law of large numbers (weak and strong), central limit theorem, hypothesis testing. [5 Lectures]
- **Module VII:** Concentration inequalities Chernoffs bound, Hoeffdings inequality and their applications in parameter estimation and confidence interval of parameters. [3 Lectures]
- **Module VIII:** Random vectors and covariance matrix. Random processes. Autocorrelation, cross correlation, power spectral density. Basic notion of ergodicity. [6 Lectures]

Text books:

1. Grimmett and Stirzaker, Probability and Random Processes, 4th Edition, Oxford University Press, 2020.
2. Papoulis and Pillai, Probability, Random variables and Stochastic processes, 4th Edition, McGrawHill Europe, 2002.

References

1. Erhan Cinlar, **Introduction to Stochastic Processes**, Dover Books on Mathematics, 2013.
2. R. G. Gallager, **Stochastic Processes: Theory for applications**, Cambridge University Press, 2014.
3. S. M. Ross, **Stochastic processes**, 2nd Edition, John Wiley, New York, 1996.
4. J. R. Norris, **Markov chains**, Cambridge University Press, Cambridge, 1999.
5. Joseph K. Blitzstein and Jessica Hwang, **Introduction to Probability**, CRC Press.com
6. Kishor S Trivedi, **Probability & Statistics with Reliability, Queuing, and Computer Science Applications**, Prentice Hall.

It is a subset of the existing course EE534 with topics relevant for MTech CSE. The 2 credit structure for this course is already approved in the senate document for CSE MTech.